Lagnange Multiplier (Max/Min)

max/Min $z = f(x,y)$

Constraint to $g_1(x,y) = k$

$g_1(x,y) - k = 0$

$g(x,y) = 0$

$\vec{\nabla} \rightarrow$ nabla

$\vec{\nabla} \rightarrow \frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k}$

$$\vec{\nabla} f = \frac{\partial f}{\partial x}\hat{i} + \frac{\partial f}{\partial y}\hat{j} + \frac{\partial f}{\partial z}\hat{k}.$$

$$\boxed{\vec{\nabla} f = \lambda \vec{\nabla} g} \rightarrow \text{Solve problems}$$

$\lambda \rightarrow (x_0, y_0, z_0) / (x_0, y_0)$

Ex: Use Lagrange Multipliers to find the maximum & minimum values of f subject to the given constraint. Also find the points at which these extreme values occur

$$\boxed{f(x,y) = 4x^3 + y^2}$$

subject to $\boxed{2x^2+y^2=1}$

Sol:

$f(x,y) = 4x^3+y^2$, $\quad \vec{\nabla} f = \frac{\partial 4x^3}{\partial x}\hat{i} + \frac{\partial y^2}{\partial y}\hat{j}$

$2x^2+y^2-1 = 0 = g(x,y) \quad = 12x^2\hat{i} + 2y\hat{j}$

So, $g(x,y) = 2x^2+y^2-1$; $\quad \vec{\nabla} g = 4x\hat{i} + 2y\hat{j}$

$$\vec{\nabla} f = \lambda \vec{\nabla} g$$

$$\Rightarrow 12x^2\hat{i} + 2y\hat{j} = \lambda\left(4x\hat{i} + 2y\hat{j}\right).$$

$$\Rightarrow 12x^2\hat{i} + 2y\hat{j} = 4\lambda x\hat{i} + 2y\lambda\hat{j} \quad \cdots\cdots (i)$$

Equating coefficients from (i)

$12x^2 = 4\lambda x$	$2y = 2y\lambda$
$\Rightarrow 3x^2 = x$	$\Rightarrow 2y\lambda - 2y = 0$
$\Rightarrow x - 3x^2 = 0$	$\Rightarrow y(\lambda-1) = 0$
$\Rightarrow x(1-3x) = 0$	$\Rightarrow y = 0,\ \lambda = 1$
$x = 0,\ x = 1/3$	

$x=0,$

$2x^2+y^2=1$

$\Rightarrow y^2=1$

$\Rightarrow y=\pm 1$

$(0,1),(0,-1)$

$x=1/3$

$2(1/3)^2+y^2=1$

$\Rightarrow \frac{2}{9}+y^2=1$

$\Rightarrow y^2=1-\frac{2}{9}$

$\Rightarrow y^2=\frac{7}{9}$

$\Rightarrow y=\pm\frac{\sqrt{7}}{3}$

$\left(1/3, \frac{\sqrt{7}}{3}\right), \left(1/3, -\frac{\sqrt{7}}{3}\right)$

$y=0$

$2x^2+0=1$

$\Rightarrow x=\pm\frac{1}{\sqrt{2}}$

$\left(1/\sqrt{2}, 0\right)\ \left(-\frac{1}{\sqrt{2}}, 0\right)$

$f(x,y)=4x^3+y^2$

(x,y)	$(0,1)$	$(0,-1)$	$\left(1/3, \frac{\sqrt{7}}{3}\right)$	$\left(\frac{1}{3}, -\frac{\sqrt{7}}{3}\right)$	$\left(\frac{1}{\sqrt{2}}, 0\right)$	$\left(-1/\sqrt{2}, 0\right)$
$f(x,y)=4x^3+y^2$	1	1	$4\cdot\frac{1}{27}+\frac{7}{9}$ $=0.93$	$4\cdot\frac{1}{27}+\frac{7}{9}$ 0.93	$4\left(\frac{1}{\sqrt{2}}\right)^3$ $\sqrt{2}$ 1.41	$-4\left(\frac{1}{\sqrt{2}}\right)^3$ $-\sqrt{2}$ -1.41

Maximum: $f(x,y)$ has maximum value $\sqrt{2}$ at $\left(\frac{1}{\sqrt{2}}, 0\right)$.

Minimum: $f(x,y)$ has minimum value $-\sqrt{2}$ at $\left(-\frac{1}{\sqrt{2}}, 0\right)$.

$$f(x,y,z) = 2x+y-2z \qquad \vec{\nabla} f = \frac{\partial}{\partial x}(2x+y-2z)\hat{i} + \cdots$$

subject to $\boxed{x^2+y^2+z^2=4}$

$$\Rightarrow x^2+y^2+z^2-4=0=g(x,y,z)$$

$$\vec{\nabla} f = \lambda \vec{\nabla} g$$

$$\Rightarrow \frac{\partial}{\partial x}2x\,\hat{i} + \frac{\partial}{\partial y}y\,\hat{j} + \frac{\partial}{\partial z}(-2z)\,\hat{k} = \lambda\left(\frac{\partial}{\partial x}x^2\,\hat{i} + \frac{\partial}{\partial y}y^2\,\hat{j} + \frac{\partial}{\partial z}z^2\,\hat{k}\right)$$

$$\Rightarrow 2\hat{i} + \hat{j} - 2\hat{k} = \lambda\left(2x\hat{i} + 2y\hat{j} + 2z\hat{k}\right) \quad \cdots (1)$$

$$2x\lambda = 2, \quad 2y\lambda = 1, \quad 2z\lambda = -2$$

$$x\lambda = 1, \quad y\lambda = \frac{1}{2}, \quad z\lambda = -1$$

$$\Rightarrow \lambda = \frac{1}{x}, \quad \lambda = \frac{1}{2y}, \quad \lambda = -1/z$$

$$\frac{1}{x} = \frac{1}{2y} = -1/z = \lambda$$

$$\Rightarrow \frac{1}{x} = \frac{1}{2y}$$

$$\Rightarrow 2y = x$$

$$\Rightarrow y = \frac{x}{2}$$

$$x^2 + y^2 + z^2 = 4$$

$$\Rightarrow x^2 + \frac{x^2}{4} + x^2 = 4$$

$$\Rightarrow \frac{4x^2 + x^2 + 4x^2}{4} = 4$$

$$\Rightarrow -1/z = \frac{1}{x}$$

$$\Rightarrow z = -x$$

$$\Rightarrow 9x^2 = 16$$

$$\Rightarrow x^2 = \frac{16}{9}$$

$$\Rightarrow x = \pm \frac{4}{3}$$

$$y = \pm \frac{4}{6} = \pm \frac{2}{3}$$

$$z = \mp \frac{4}{3}$$

$$\left(\frac{4}{3}, \frac{2}{3}, -\frac{4}{3}\right)$$

$$f\left(\frac{4}{3}, \frac{2}{3}, -\frac{4}{3}\right) = 2\left(\frac{4}{3}\right) + \frac{2}{3} - 2\left(-\frac{4}{3}\right)$$

$$= 6 \rightarrow \text{max}$$

$$\left(-\frac{4}{3}, -\frac{2}{3}, +\frac{4}{3}\right)$$

$$f\left(-\frac{4}{3}, -\frac{2}{3}, \frac{4}{3}\right) = 2\left(-\frac{4}{3}\right) - \frac{2}{3} - 2\left(\frac{4}{3}\right)$$

$$= -6 \rightarrow \text{min}$$